

Citizen Grievance System

Simple team handbook for a confident panel presentation

One-line pitch

We help citizens report civic problems quickly. Our AI sends each complaint to the right department, gives it a priority score, finds likely duplicates, and lets officials act only inside their own department and region.

Problem statement AI-Based Citizen Grievance Classification, Prioritization and Duplicate Complaint Detection	Project code SIH26-S02
Demo stack React + Express + MongoDB + FastAPI AI	Current local data Persistent MongoDB: citizencare

Read this once before the presentation. It explains what to say, what to show, and how to answer common questions.

1. The project in one minute

Say this to the panel

“A citizen submits a civic issue such as a pothole, garbage pile, broken street light, water leak, or public safety risk. Our AI reads the complaint, identifies the department, assigns a 1-to-5 priority, and checks if a similar complaint already exists nearby. The complaint is saved in MongoDB, officials can update its status, and citizens can track the result.”

Why this matters

- Citizens do not need to know which government department handles the issue.
- Urgent risks, such as a live wire or gas leak, are not treated like routine complaints.
- Duplicate reports are kept as evidence of public impact instead of being silently deleted.
- Officials see only the work assigned to their department and region.
- The system gives administrators one place to monitor service quality and workload.

The core flow

1. Citizen reports Title, description, optional photo and location	2. AI analyses Department, priority, duplicate match	3. MongoDB saves Permanent tracking record	4. Official acts Status update and note
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2. What is special about our solution

The system is not only a form for complaints. It uses practical AI and safe access control.

Department routing

The AI maps a complaint to a civic department, for example Public Works, Sanitation, Water Supply, Electricity, or Public Safety.

Priority scoring

Each complaint gets a score from 1 to 5. Higher scores represent more urgent situations.

Duplicate detection

The system compares the new complaint with past complaints using semantic similarity and location. A similar pothole 40 km away is not called a duplicate.

Region and department security

An official is assigned a department and state/UT. Their queue, statistics, analytics, and updates are limited to that exact scope.

Citizen trust

Officials can update status and add notes, but cannot rewrite or delete what a citizen originally reported.

Graceful fallback

If an external AI service is unavailable, the complaint is still saved using a deterministic rules engine.

Easy way to explain duplicate detection

“We compare meaning, not only exact words. ‘Large pothole on MG Road’ and ‘Huge crater on M.G. Road’ can be recognised as the same issue if they are close by. But we keep both reports so the original issue gets an upvote and shows its real public impact.”

3. How the system works technically

Use this section when the panel asks about architecture. Keep the explanation simple first, then add detail only if asked.

React portal Citizen, official, owner, map, feed and analytics screens.	Express API Authentication, permissions, issues, votes, SLA and business rules.
FastAPI AI service Department routing, priority score and semantic duplicate detection.	MongoDB Persistent system of record for users and grievances.

Important technical points

- MongoDB is the source of truth. The vector index is only used to help find similar complaints.
- The AI service is called after the citizen submits a complaint. If it is temporarily unavailable, the complaint is not lost.
- The duplicate index is updated only after the complaint has been saved in MongoDB. This avoids pointing to a record that does not exist.
- The current local database is persistent. Closing the app or terminal does not remove its data.

AI method in simple words

We use layered analysis. Keyword rules always give a fast answer. An optional language model can add context. A validation layer stops the model from inventing departments or giving a dangerous complaint a low priority. For duplicate detection, sentence embeddings plus FAISS compare the meaning of complaints, and a location check prevents false matches across distant places.

4. Roles and the demo plan

Role	What to demonstrate	Safe talking point
Citizen	Register/sign in, file complaint, view My Issues, feed and map.	The citizen can report and track; they cannot change government data.
Official	Sign in at /official, open analytics, change only a valid issue status.	Official sees only the assigned department and region.
Owner admin	Open /owner, show AI health, staff access, summary counts and dashboard.	Owner has system-wide operational access.

Suggested 4-minute demo

- **0:00-0:30** Open the landing page. State the one-line pitch.
- **0:30-1:30** Show a citizen complaint. Explain AI routing, priority and the location-aware duplicate check.
- **1:30-2:20** Sign in as an official. Show that the queue and analytics are limited to Public Works Department - Karnataka.
- **2:20-3:10** Update a complaint from Pending to In Progress and add a note. Explain that officials cannot edit or delete citizen content.
- **3:10-4:00** Open owner/admin view and MongoDB Compass or mongosh. Show the stored `issues` and `users` collections.

Local demo accounts

All demo accounts use the password **demo12345**. Citizen: citizen@demo.in. Official: officer@gov.in. Owner: admin@demo.in. These are local demonstration accounts only.

5. Showing the permanent database to the panel

The database is local and persistent. Its name is **citizencare**. It stores the real app records used in the demo.

What to show

- **issues:** title, department, state, status, priority score, duplicate information and AI analysis.
- **users:** name, email, role, department and region. Password hashes are not returned to the UI.
- Explain that raw Aadhaar numbers are never stored. The system keeps only verification status and time.

Option A: MongoDB Compass (best visual demo)

Open MongoDB Compass and connect using: `mongodb://localhost:27017/citizencare`. Then open ``issues`` and ``users``. You can filter with ``{ department: "Public Works Department", state: "Karnataka" }``.

Option B: Terminal

Commands to run

```
mongosh mongodb://localhost:27017/citizencare

show collections
db.issues.find({}, {title:1, department:1, state:1, status:1,
priorityScore:1}).limit(20)
db.users.find({}, {name:1, email:1, role:1, department:1, region:1})
```

Useful one-line answer

“MongoDB is our persistent system of record. Every submitted grievance and every account role is stored there. The AI index helps search for duplicates, but MongoDB remains the trusted database.”

6. Common panel questions and easy answers

Q. Why do you need AI?

A. Citizens may not know the correct department or urgency. AI helps route and prioritise quickly. The final system still stores the complaint even if AI is unavailable.

Q. How do you stop wrong AI decisions?

A. We use rules first, optional language-model help second, and a validation layer last. Hazard keywords can override a weaker infrastructure match. The model cannot invent a new department.

Q. How do you detect duplicates?

A. We compare complaint meaning using embeddings and check distance. A similar complaint nearby can be linked; identical text far away is treated as a separate issue.

Q. Do you delete duplicate complaints?

A. No. We save them, give the citizen a tracking record, and add support to the original issue. This shows how many people are affected.

Q. How is official access controlled?

A. The owner assigns every official a department and region. The server filters queues, analytics and updates on every request, so changing the browser cannot expose other data.

Q. What is stored for Aadhaar?

A. Only whether the verification step succeeded and when. The raw Aadhaar number is never stored, logged or returned.

Q. What happens after a restart?

A. The app data stays in the local persistent MongoDB service. Only the temporary in-memory demo mode wipes its records when stopped.

Q. What is a current limitation?

A. Duplicate quality depends on the embedding model and language used. A multilingual model can improve code-switched Indian language matching. The local service is for demo/development, not a public production deployment.

Final reminder for the team

Do not try to explain every library. Start with the citizen benefit, then show the flow, then explain AI, security and MongoDB only when asked. Be honest about current limits; that builds confidence.